

Understanding and Designing Technologies for Family Health Routines: Supporting Children in the Digital Age

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ABSTRACT

Child health and well-being are declining in the U.S. with decreases in physical activity and preventive medical visits [19] and increases in child anxiety and depression, and especially since COVID-19. Children's health-related routines are developed and maintained within families to ensure that health-enhancing behaviors occur. In middle childhood, children have the capacity to be involved in tracking and planning routine health behaviors. However, technologies for family routines do not typically involve children's input. Additionally, children, parents, and researchers disagree on the extent to which technology should be a part of children's routines. Thus, it is important to study the role of technology in children's routines and the extent to which they want autonomy support. I apply a theoretical perspective that frames family routines as a form of collective behavior that renders health-related behaviors automatic, and that rely upon artifacts to prompt such automatic behaviors (e.g., teeth brushing). The dissertation uses an interview study and participatory design sessions with child-parent dyads to explore the nature of children's health-related routines in the digital age to better understand: (1) the role, if any, technology plays in family routine development, maintenance, and performance, (2) the impact of these uses on child health and well-being, (3) how parenting style influences these uses, (4) how children want technology to support their health and well-being, and (5) the extent to which children want technology to support autonomy in the development, maintenance, and performance of family routines. This dissertation will inform the design of family informatics tools that can provide autonomy support, which can positively influence child health and well-being.

KEYWORDS

late childhood, early adolescence, caregivers, routines

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1 INTRODUCTION

Children's health-related routines (e.g., meals, physical activity) are critical contributors to their health, well-being, and development [6, 7]. Middle childhood is a crucial time for healthy routines to be developed as that is when health information is absorbed, health attitudes are formed, and foundational health practices are initiated [6]. Caregivers are the "gatekeepers" to family health [33], as they help children create and adhere to healthy routines, as in those routines that include positive health behaviors [i.e., choices, actions, or practices that could impact health outcomes [3]].

Family routines are "the setting" in which the health of individual family members is developed and sustained [7]. Family routines are observable, predictable, and repetitive behaviors that occur daily or weekly to help maintain the structure and emotional climate of family life [8, 35, 38]. Family routines can be unplanned or intentional, deliberate, and directed by caregiver values and predictable life events [6]. Family routines facilitate collaboration across performance of health routines, and collaboration is vital to the sustainability of health routines because, for many people, family members are emotionally invested [7] and can be a source of encouragement in which the members reciprocate and help one another [4, 12, 25], even amidst disruption [1, 31].

The household is a place in which technologies play a major role in family routines, since artifacts support coordination of home life [15]. Given the pervasiveness of technology in family life, my research interests begin with three expected roles that technology can play in healthy family routines. First, use of technology can be the routine, as in using mindfulness apps. Second, technology can be an intervention that facilitates routine behavior, as in the use of calendars or reminders. Third, technology can serve both functions to monitor and facilitate technological routines, keep children on a schedule, and serve as a forcing function to stick with a plan, such as through screen time management.

The COVID-19 pandemic effectively changed life at home, as technology use increased by adults and children due to the frequency of both hybrid schooling and work. This adverse event required some parents to change their role in their children's daily life [26]. Realizing the potential impact of technology use on child mental health, some parents tried to intervene in their children's technology routines, using monitoring technology or by establishing family technology rules [9]. Yet, parenting styles have not been considered in the study of technological intervention in family health-related routines (i.e., online and offline). Thus, I am interested in the role of technology in children's routines and the extent to which parenting style influences use.

Family informatics is a human-centered computing science focused on the design, deployment, and evaluation of human-facing

technological solutions to promote and manage family behaviors, as individuals and as a whole [25]. Scholars have also developed technological interventions to support families as a unit and applied technology to specific behaviors [14, 29]. While Piña et al. [25] describe family informatics in the context of family behaviors broadly, the definition has not been used to study cognitive or social-emotional well-being; thus, I extend the definition of family informatics to also cover cognitive and social-emotional health which are intertwined in child health and well-being.

Family informatics scholars have also highlighted some challenges of information sharing behaviors within families due to privacy concerns [12, 31], surveillance, and violation of trust [41]. At times, children feel misunderstood regarding the educational value of technology use (e.g., gaming) [10], and conflict ensues between children and caregivers [17, 22]. The lack of shared understanding between parents and children on the purpose of technology routines contributes to parent-children tension. In building on this work, I aim to investigate children and parents' perspectives on family routine, and how design can support achieving a shared understanding to which children contribute as partners.

Finally, most technological interventions that support children's healthy behavior-specific routines (e.g., sleep, eating, play) involve caregivers as primary users, do not engage or request input or feedback from children, and do not support co-use by caregivers and children together [21, 39]. However, a handful of studies have acknowledged children's capacity to be involved in tracking and planning routine health behaviors [14, 25, 32]. Thus, there is an opportunity for me to specifically investigate children's interest and ability (i.e., cognitive and social emotional capacity) to take a more active role in the facilitation and maintenance of routines and study how technology might facilitate this. I will particularly focus on behaviors that positively contribute to children's health and well-being (e.g., sleep, eating, play, and relaxation).

2 DISSERTATION OUTLINE

I will use a multi-phase approach that considers perspectives of parents and other caregivers, children, the family as a whole, and the home environment simultaneously to explore how children and families develop, maintain, and perform their children's health-related routines.

Towards this goal, I will collaborate with both children and their parents/guardians to explore where children want to/can obtain more autonomy in their health-related routines, the extent to which parenting style affects this process, and how informatics interventions might support children's autonomy. I will use an interview study and participatory design sessions to answer the following research questions: (1) What role does technology play in the development of child and family health-related routines?, (2) How do children and parents/guardians use technology to develop and perform child and family health-related routines? Does parenting style have a relationship with these uses of technology? How much autonomy do children have in these uses?, (3) How do technology-mediated child and family routines influence child health behaviors? How would children like to use technology to develop and perform their health-related routines?, and (4) How

might technology build on family routines to positively support child health and well-being?

3 AIM 1 METHODOLOGY

To characterize, and identify challenges to, child health and well-being routines, and uncover the role of technology in such routines, I conducted 33 semi-structured interviews with children (ages 7-12) and their parents/grandparents. Each child and parent/grandparent participant also completed a survey. The interviews explored family routines with special interest in the role of technology in the development and maintenance of their health-related routines, so as to understand the role of technology.

3.1 Sampling and Recruitment

Parent or guardian participants needed to be at least 18 years old and children needed to be between 7 and 12 years old. I recruited 33 families to participate in the study. I determined sample sizes for the in-depth interviews based on previous recommendations suggesting that thematic saturation in purposive sampling may be achieved with approximately ten to twelve interviews in a homogeneous group, but larger numbers in heterogeneous samples [11, 13, 18]. I employed a purposive sampling [11] strategy to maximize diversity of ethnicity, race, child age and caregiver role to maximize variability in family perspectives.

I applied strategies in Metro Detroit area to recruit participants, including 1) the University Health Research Website, 2) social media (e.g., Facebook groups, NextDoor), and 3) virtual and paper flyers/newsletter entries to family-focused organizations (e.g., YMCA, churches, sports centers). This strategy and an incentive (\$50) helped increase participation amongst families who might not otherwise be involved or seek research participation opportunities. Study candidates were invited to fill out a 5-minute survey online (i.e., Qualtrics) to determine eligibility. I used snowball sampling, asking interview participants to invite others who qualify for this study. Recruitment stopped when the target number of participants per category was achieved.

This sampling strategy reached theoretical saturation across family structure type (i.e., nuclear family, single parent family, step-family/blended family, and extended family), SES (i.e., low, middle, and high), and child age (i.e., late childhood and early adolescence). Thus, the proposed sample is suited to identify main themes across narratives and allow for comparisons across socioeconomic status and family structure.

3.2 Data Collection

Prior to interviews, I sent parents/guardians a Qualtrics survey to obtain demographics to help describe the sample and assess parenting styles using the Parenting Styles and Dimensions Questionnaire-Short (PSDQ [28]) and the Family Routines Inventory [16].

During the interview, I asked the child about their weekly routines mentioned by the parent from the survey, as well as about which of those routines they consider healthy. This helped me develop a baseline understanding of their definition of healthy routines and then provide the opportunity for me to ask children and caregivers questions in connection with those routines.

At the end of the interview, I asked children to fill out a survey that includes basic demographic questions, as well as the following items from the following survey measures: Revised Mental Health Inventory-5 (MHI-5 [27]), the Child or Adolescent Routines Questionnaires (CRQ, ARQ) [23, 36–38], Youth Physical Activity Questionnaire (YPAQ [5]), and items from questionnaires on Sleep and Screen Time [34] and child-parent conflict [24].

3.3 Data Analysis

I wrote contact summaries after each interview and then analytic memos to explore and reflect on the data and take note of remaining questions [30]. In weekly meetings with my advisor, I continue to discuss my interpretations and memos and continue developing codes by comparing findings across all families observed and family members interviewed. I will upload interview data into NVivo for analysis. Using NVivo, I will initially analyze transcripts inductively using an open coding approach [30] to identify and stay open to changing themes as I analyze my data, as well as a deductive approach in connection with my definitions of routines to build on the literature.

I will iteratively review the themes found in the data against topics of interest, such as parents' values that inform the development of healthy routines and perspectives on the involvement of technology in healthy activities. In an effort to compare interview data, I will continue to code the data and practice memoing, and discuss the codes in weekly meetings to generate, review, define, and finalize themes. I will also use informal member checking [20] with local family partners to confirm that my themes align with family experiences. My findings will ultimately be constructed through comparisons to the literature and across my data.

3.4 Expected Contribution

This work will add to the informatics literature by describing the role of technology in family routines and their potential for influencing health, as well as children's preference for how technological intervention may support their routines.

4 AIM 2

Aim 1 findings motivate the direction of Aim 2, which will take a more participatory approach to ideation and design. A 2020 review of digital mental health interventions for children [2] highlighted the need for research with ethnically and socio-economically diverse groups of younger children (< 13 years old) that is also relevant to caregivers who may support its use (e.g. parents, family members, and teachers). This work also outlined the lack of participatory research methods used to obtain the insights of these populations, which motivated my methodological choice.

Empirically, Aim 2 will approach family routines through participatory design with children and their parents/guardians. I will use design activities to encourage children to design routines, and potential technological infrastructures, that are resilient to disruptions and able to meet children's desire for autonomy in the maintenance of their routines as per Aim 1. Such design will create opportunities for conversations between child-caregiver dyads on how they would like healthy routines to be supported by technology. I may use open-ended scenarios of word association exercises to help

uncover how children and home caregiver attitudes and mental models may differ regarding children's autonomy regarding healthy routines. Dyads will ideate on solutions to known challenges to children's health-related routines (such as disruption or other examples presented by participants in Aim 1); and child/family attitudes toward the role of technology in such routines. I will ask child participants to review a list of challenges to child health-related routines, and then add to and edit the list in conversation with their caregiver and reflect on this with the broader group. I will then ask participants to design solutions to a challenge that resonates with them. I will analyze this data to understand how socio-technical solutions might better consider child and family attitudes toward the role of technology in health-related routines.

Finally, to understand how families envision technological intervention in healthy routines, I will bring one technological prototype that is supportive of children's routines, resilient to disruptions and able to meet children's desire for more autonomy in the development and maintenance of their routines. I will ask children questions about this prototype and use approaches from co-design such as layered elaboration [40] for children to iterate upon or change the existing idea.

4.1 Expected Contribution

This phase of the research will uncover how technology might support children's health and well-being routines through the provision of autonomy support, and contribute design implications for technology to provide autonomy support to children of the digital age.

5 CONCLUSION

This dissertation will characterize how families develop routines around healthy behaviors and the role of technology in such routines, and uncover the extent to which children want increased autonomy as part of their health-related routines. In future work, I aim to use participatory methods to design, and iterate on, a technology probe with child-parent dyads to understand how they use it and in what ways it might be flexible.

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